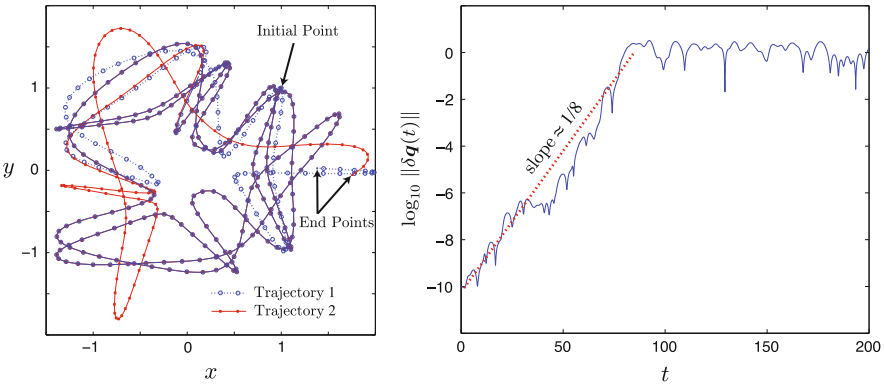



**Fig. 1.25** Rapid separation of trajectories in the perturbed central force problem. The trajectories in the *left panel* are started from initial conditions differing only in the 10th digit. They are then found to separate rapidly and end up following entirely different paths. The *right panel* shows that the trajectories separate exponentially fast in the early going; eventually, once the difference is of the same order as the size of the domain, the norm of the difference stabilizes

### *1.7.2 Variational Equations and Lyapunov Exponents*

One way of quantifying the sensitive dependence on initial conditions in a nonlinear dynamics model is via *Lyapunov exponents*. Usually this is done by introducing the *variational equations* which describe the time-dependent variation of perturbations of a solution of a dynamical system. Besides giving us a means to verify the chaotic nature of given system, the variational equations prove useful in describing the *symplectic structure* (taken up in the next chapter) which is essential to the design of effective numerical methods for molecular dynamics.

Let a dynamical system $\dot{z} = f(z)$ be given in $\mathbb{R}^m$ with flow map $\mathscr{F}_t : \mathbb{R}^m \to \mathbb{R}^m$ which we assume to be continuously differentiable. Let $z(t, \zeta)$ represent the solution of initial value problem

$$\dot{z} = f(z), \qquad z(0) = \zeta .$$

We then compute the $m \times m$ Jacobian matrix of $\mathscr{F}_t(z(t, \zeta))$:

$$W(t) = \mathscr{F}_t'(z(t, \zeta)) = \frac{\partial \mathscr{F}_t}{\partial z}(z(t, \zeta)).$$

Differentiating $W(t)$ with respect to $t$ and using the differential equation and the chain rule, we have

$$\frac{\mathrm{d}}{\mathrm{d}t} W(t) = f'(z(t, \zeta)) W(t). \tag{1.10}$$